

Declaration of Academic Integrity

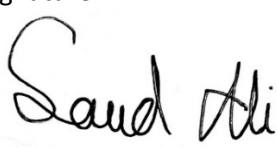
Academic Integrity is a central tenet of Singapore Polytechnic. The polytechnic rules state that "Cheating in examinations and other assessed work is a very serious offence. This includes copying and using plagiarised material. Any student who cheats, attempts to cheat or breaches any rules for examinations and tests will face disciplinary action. The student is liable to be expelled."

Check **only one** of the two options below:

☐ I affirm that the work I submit is my own, produced without help from any AI tool(s) and/or other source(s).

☒ I affirm that the work I submit has been produced with the use of AI tool(s) and/or other source(s) which I have acknowledged fully in the [following section](#).

By signing this form, I declare that the above affirmation made is true, and that I have read and understood the rules stated in Students Handbook on "[Plagiarism](#)" and "[Breach of Examination/Assessment Rules](#)".

Name: Tharik Parvas S/O Checkoussain Syed Saud Ali	Student ID: 2349288 2209111	Class: DIT/FT/2A/03
Signature: 	Date: 1/8/2024	

Acknowledgement of use of AI tool(s) and/or other source(s) (where applicable)

AI tool(s)

Name of AI tool < For example, ChatGPT >	< ChatGPT >
Input prompt < Insert the question that you asked ChatGPT >	<1. StudentAdminPanel.java File SearchAdminPanel.java File why is my NamePanel Null? > <2. StudentAdminPanel.java File This is in my student.txt file before i run the program:3 DIT/FT/2A/02;p1234567;John Tan;2;ST0509;Java;4;80;ST0502;FOP;6;75; DAAA/FT/2A/03;p2222222;Peter Goh;4;ST0501;FED;5;90;ST0502;FOP;6;78;ST2413;FOC;4;65.5;SP108B;ECG;4;81; DIT/FT/2A/04;p3333333;Jack Lim;3;ST0503;BED;6;55.5;ST0504;MAD;5;86;ST0502;FOP;6;66;. This is my student.txt file after i add 2 new student:5 DIT/FT/2A/02;p1234567;John Tan;2;ST0509;Java;4;80;ST0502;FOP;6;75; DAAA/FT/2A/03;p2222222;Peter Goh;4;ST0501;FED;5;90;ST0502;FOP;6;78;ST2413;FOC;4;65.5;SP108B;ECG;4;81; DIT/FT/2A/04;p3333333;Jack Lim;3;ST0503;BED;6;55.5;ST0504;MAD;5;86;ST0502;FOP;6;66; DIT/FT/2A/03;p2209111;Saud Ali;0; DIT/FT/2A/03;p2349288;Tharik Parvas;0;. Why is there a empty line in my txt file for the second student. >
Date generated <Insert the date that ChatGPT response was generated, since ChatGPT is an evolving technology >	<1. 28/7/2024> <2. 28/7/2024>
Output generated < Insert the response verbatim from ChatGPT >	<1. To determine why `NamePanel` is returning null when accessed through `getStudentNamePanel()`, I will review the latest code of `StudentAdminPanel.java` and `SearchAdminPanel.java`. I'll start by checking how `studentNamePanel` is instantiated and maintained in `StudentAdminPanel`, and then see how it's being accessed from `SearchAdminPanel`.

	Let's first examine the `StudentAdminPanel.java` to confirm the initialization and handling of `studentNamePanel`. The `StudentAdminPanel` code properly initializes the `studentNamePanel` along with other panels and provides a getter method `getStudentNamePanel()` to access it. This indicates that the initialization in `StudentAdminPanel` is correct. Now, let's review the `SearchAdminPanel.java` to confirm how it accesses the `studentNamePanel` and if it's being referenced correctly without any issues that might lead to it being null. I'll check this file now. The review of the `SearchAdminPanel` code indicates that it does not correctly initialize or pass a reference to an instance of `StudentAdminPanel`. The constructor of `SearchAdminPanel` does not take any parameters, which means it never receives a reference to `StudentAdminPanel`. Consequently, when it tries to access `getStudentNamePanel()`, `StudentAdminPanel` is still null, leading to a `NullPointerException`. ### Solution: Modify the Constructor and Ensure Proper Initialization **1. Modify the Constructor to Accept `StudentAdminPanel`:** Update the constructor of `SearchAdminPanel` to accept an instance of `StudentAdminPanel` and use this instance to access the panels. <pre> `java public class SearchAdminPanel extends JPanel { private StudentAdminPanel studentAdminPanel; // Correctly named variable public SearchAdminPanel(StudentAdminPanel studentAdminPanel) { this.studentAdminPanel = studentAdminPanel; // Store the passed reference // Existing setup code setBorder(BorderFactory.createTitledBorder("Search")); setLayout(new BorderLayout()); setPreferredSize(new Dimension(300, 200)); // Initialize components initComponents(); </pre>
--	--

```

    }

    private void initializeComponents() {
        searchByAdminNo = new JRadioButton("By Admin");
        searchByName = new JRadioButton("By Name");
        ButtonGroup searchGroup = new ButtonGroup();
        searchGroup.add(searchByAdminNo);
        searchGroup.add(searchByName);
        searchField = new JTextField(10);
        searchButton = new JButton("Search");

        // Add components and set up listeners
        add(searchByAdminNo, BorderLayout.NORTH);
        add(searchByName, BorderLayout.CENTER);
        add(searchField, BorderLayout.SOUTH);
        add(searchButton, BorderLayout.SOUTH);

        searchButton.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                // Usage of studentAdminPanel
                try {
                    LabelledInputAdminPanel namePanel =
studentAdminPanel.getStudentNamePanel();
                    System.out.println("Name Panel Text: " + namePanel.getText());
                } catch (NullPointerException npe) {
                    System.out.println("StudentAdminPanel or NamePanel is null.");
                    npe.printStackTrace();
                }
            }
        });
    }
}
...

**2. Ensure Proper Instantiation in Main Application:**
When creating an instance of `SearchAdminPanel`, make sure you pass the
already created and initialized `StudentAdminPanel`:

```java

```

```

public class Main {
 public static void main(String[] args) {
 JFrame frame = new JFrame("Application");
 frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

 StudentAdminPanel studentPanel = new StudentAdminPanel();
 SearchAdminPanel searchPanel = new SearchAdminPanel(studentPanel);

 frame.add(studentPanel, BorderLayout.NORTH);
 frame.add(searchPanel, BorderLayout.CENTER);
 frame.pack();
 frame.setVisible(true);
 }
}
...

```

By following these steps, you ensure that `SearchAdminPanel` correctly receives and maintains a reference to `StudentAdminPanel`, allowing safe access to `studentNamePanel` without encountering null references. If you continue to encounter issues, please double-check all connections and initialization sequences in your application setup.>

<2. Since commenting out `TotalStudent` resolves the issue of extra lines, the issue likely lies within how the `TotalStudent` method handles file modifications. Let's examine the relevant part of the `TotalStudent` method to identify potential issues:

```

```java
public void TotalStudent(boolean isIncrement) {
    if (isIncrement) {
        try {
            String file = "student.txt";
            Scanner sc = new Scanner(new File(file));
            StringBuffer buffer = new StringBuffer();
            while (sc.hasNextLine()) {
                buffer.append(sc.nextLine() + System.lineSeparator());
            }
            String fileContents = buffer.toString();
            String[] lines = fileContents.split(System.lineSeparator());
            String OutDatedStudentNo = lines[0];
            int oldTotal = Integer.parseInt(OutDatedStudentNo.trim());

```

	<pre> int NewTotal = oldTotal + 1; String UpdatedStudentNo = String.valueOf(NewTotal); fileContents = fileContents.replaceFirst(String.valueOf(oldTotal), UpdatedStudentNo); FileWriter writer = new FileWriter(file); writer.append(fileContents); writer.close(); } catch (Exception e) { System.out.println("Exception: " + e); } } else { // Code to decrease total by 1 } } ... </pre> ### Potential Issue **Appending Whole File Content**: When updating the file contents, the method reads the entire file into a <code>StringBuffer</code>, modifies it, and then writes it back using a <code>FileWriter</code>. The usage of <code>System.lineSeparator()</code> when appending each line back into the <code>StringBuffer</code> can introduce extra new lines, especially if the original file already includes newline characters at the end of the file. **Last Line Newline Check**: Before appending the updated contents back to the file, it does not check whether the last line in <code>buffer</code> already ends with a newline. This could lead to doubling up on newlines, especially if <code>fileContents</code> had newline characters at its end. ### Solution Suggestion To address this, you can adjust how you handle the reading and writing process to avoid unnecessarily appending new lines: **Check Last Character**: Modify the <code>TotalStudent</code> method to check if the last character of the buffer is a newline before appending another. **Control Newline Addition**: When constructing the <code>StringBuffer</code>, control how you add newline characters, especially after the last line read from the file. Here's a suggested modification:
--	--

	<pre> '''java while (sc.hasNextLine()) { String line = sc.nextLine(); buffer.append(line); if (sc.hasNextLine()) { // Only append a newline if there are more lines to read buffer.append(System.lineSeparator()); } } ''' </pre> This approach ensures that you only add newlines when they are necessary (i.e., between lines), preventing an extra newline at the end of the file. This should help avoid unnecessary blank lines due to `TotalStudent` manipulations.>
Impact on submission < Briefly explain which part of your submitted work was ChatGPT's response applied >	<1. I was able to get the NamePanels from another file in the file SearchAdminPanel. The changes I made is with MainFrameAdmin, as I needed to include the studentpanel parameter inside the searchpanel to access it.> <2. I was struggling to find out why does it create extra lines in the txt file, so with the chatgpt suggestion being successful, I just applied it with understanding.>

Other source(s)

Source	Impact on submission
< 1. https://www.javatpoint.com/java-program-to-append-a-string-in-an-existing-file > < 2. https://www.tutorialspoint.com/how-to-overwrite-a-line-in-a-txt-file-using-java > < 3.	<1. I was able to correctly append the data to the file.> <2. I was able to overwrite the data given. Instead of finding a specific code to replace, I had it find the number in the first line of the code, extract it as an integer and increment by 1, before replacing the old number with the new and update the file respectively.> <3. I was able to read the file and see if the admin number the user inputs exist or not.>

https://www.tutorialspoint.com/how-to-find-the-strings-within-a-text-file-in-java > Link to online source	
< 1. > Author, Book / Article / Journal, Year, Page	< Briefly explain which part of your submitted work the Book / Article / Journal was applied >
< 1. > <i>Student / Person name</i>	< <i>Briefly explain which part of your submitted work the help of another student / person was applied</i> >
< 1. > <i>Others</i>	< <i>Briefly explain which part of your submitted work the source was applied</i> >